Environmental Implication

PFAS are persistent, mobile and bioaccumulative contaminants, found in surface water at ng/L to low µg/L and regulated at 0.65 ng/L (EU environmental quality standard, PFOS) and 4 ng/L (US EPA maximum contaminant level). Measured aquatic toxicity exists for fewer than one hundred of over eight thousand registered PFAS, so criteria for the rest must come from models. We show that their training endpoint, the most sensitive reported effect concentration, is confounded with testing effort by up to 4.9 $\log_{10}$ units, and that the confound propagates into any criterion derived from it. We supply a diagnostic that detects it.